SOLUTION BRIEF

MaiaEdge Federated Private Networking for Distributed AI Infrastructure

Distributed AI Infrastructure.

Every GPU cluster you deploy is a networking island. You're rebuilding edge infrastructure from scratch — BGP, ASNs, public IPs, redundancy at every site, for every customer. It shouldn't be this hard.

Private paths. Any network. Instantly.

CARRIER INFRASTRUCTURE FOR FEDERATED PRIVATE NETWORKING

01 THE PROBLEM

1. The Problem: Distributed AI Infrastructure Is a Networking Nightmare

Every GPU cluster you operate runs as an isolated island. The result: your team is a networking integrator at every site, not an AI infrastructure operator.

1.1 What We Heard

These are real operational pain points from operators managing multi-site GPU clusters:

> *Every new site is a full networking project from scratch. BGP, ASNs, public IPs, redundancy — we rebuild it all, every time.*
>
> *We're not selling AI infrastructure. We're selling BGP engineering services with GPUs attached.*
>
> *Our customers want to extend their on-prem network into our cluster. That should be a product. Right now it's a six-week project.*

02 THE SOLUTION

2. How MaiaEdge Solves It

MaiaEdge replaces the traditional BGP-heavy cluster edge with a managed Ethernet fabric - the Path Border Controller (PBC 2000) at each cluster, coordinated by a cloud-based Path Computation Engine (PCE). The result: Ethernet-in, Ethernet-out, centrally managed.

2.1 The Architecture

- PBC 2000 placed at each cluster edge — replaces the BGP/ISP complexity with a managed handoff point
- Two PBCs per cluster (one per ISP, both tied into both spines) preserves your HA posture
- Cloud PCE provides centralized path selection across all clusters — cost-optimized or latency-optimized routing
- Arc Net: one or a few owned entry/exit points; Arc standardizes internet access instead of rebuilding it per site
- Customers receive Layer 2 connectivity — they use their own addressing, no BGP exposure, no ASN required

2.2 Pain to Solution Mapping

PAIN POINT	MAIAEDGE SOLUTION
BGP complexity rebuilt at every site	PBC 2000 replaces ISP BGP handoff with managed Ethernet
Customer needs on-prem extension	Layer 2 handoff — customer uses their own addressing
Eight sites, eight ISP setups	Arc Net: one or two owned entry/exit points, PCE routes all
60-90 day circuit provisioning	PCE brokers cross-operator path in seconds

3. Use Cases

3.1 Whole-Cluster Allocation to a Single Buyer

Today: you book and provision every server individually. One PE-backed buyer wanting a full cluster means weeks of per-node work.

With MaiaEdge: the cluster is presented as a single segmented environment. The buyer gets a Layer 2 handoff. You allocate in a single operation, not N server tickets.

- Immediate monetization of large-block capacity
- Buyer experience: cluster behaves like on-prem infrastructure
- No ASN, no public IP block, no BGP discussion with the customer

3.2 AI Startup: Colo GPU + Cloud Front-End

The pattern: AI startups keep inference front-end and API services in AWS or Azure. They want GPU training/inference in colo for cost and control. Bridging that cleanly is the blocker.

With MaiaEdge + Equinix integration: private path from cluster edge directly into AWS/Azure on-ramps. No VM-based router. No NAT hairpin. Low latency, deterministic routing.

- Colo GPU cluster behaves like an extension of their cloud VPC
- You offer a premium 'cloud bridge' SKU - differentiated product, not just raw iron
- Startups don't manage the connectivity - you do

3.3 Single Net: Centralized Internet Access Across All Sites

Today: every cluster has its own ISP deal, its own BGP config, its own edge setup. Eight sites means eight times the work.

With MaiaEdge: establish one or two Arc-owned entry/exit points. Standardize internet access through the PCE (Path Computation Engine) in MaiaEdge Cloud. Every new site added to Arc Net inherits the fabric - no new ISP engineering from scratch.

- New cluster onboarding time: days, not weeks
- Centralized path policy: route by cost at night, route by latency during peak inference windows
- Resale opportunity: cross-connect-style services to tenants on the fabric

3.4 Multi-Operator Federation

The longer play: Arc clusters aren't just your infrastructure — they can federate with other operators. A customer in your Montreal cluster can have deterministic private connectivity to a partner cluster in Frankfurt or Singapore.

- Sovereign routing: customer data never traverses the public internet
- Differentiated from hyperscaler: no hyperscaler lock-in, operator-controlled path
- MaiaEdge PCE brokers the cross-operator path in seconds — not 60-90 day circuit provisioning

04 REFERENCE DEPLOYMENT

4. Reference Deployment

4.1 Recommended Architecture Per Cluster

This is the pattern MaiaEdge recommends for operators with an existing dual-ISP, dual-spine design:

- Two PBC 2000 units deployed at cluster edge — one per ISP link
- Each PBC connects to both spines (cross-connected) — preserves full HA; no new SPOF

- Existing BGP session between ISP and spine is replaced by PBC-managed handoff
- PCE registered to cloud plane; cluster appears in centralized fabric view
- Customer access provisioned as Layer 2 segment — own addressing, isolated from other tenants
- Optional: ship small customer-edge device so cluster appears 'on-prem' from customer perspective

4.2 Onboarding a New Cluster

STEP	ACTION	TIME
1	Ship PBC 2000 units to site	Day 0
2	Rack, power, and cable to spines	Day 1
3	Register PBC to MaiaEdge Cloud PCE	Day 1
4	Cluster appears in centralized fabric view	Day 1
5	Provision customer Layer 2 segment	Day 2
6	Customer connects — no BGP, no ASN	Day 2

05 WHY MAIAEDGE

5. Why MaiaEdge

Existing options don't solve this.

- Traditional SD-WAN solves the edge to enterprise. It doesn't solve the middle — the cross-operator, cross-cluster path between GPU infrastructure and customers.
- Hyperscaler networking (AWS TGW, Azure vNet peering) locks you in and doesn't extend cleanly to colo.
- DIY BGP fabric is what you have today. It is complicated.

MaiaEdge is the only purpose-built federated private networking layer for distributed AI infrastructure. The PBC is purpose-designed for this edge handoff. The PCE is built for cross-operator path computation. The category is Federated Private Networking — we named it because nothing else fit.

06 CONCLUSION

6. Conclusion

Distributed AI doesn't have a compute problem. It has a networking problem. Every cluster you deploy today is a manual rebuild - BGP, ISPs, public IPs, per-node provisioning, repeated from scratch, at every site, forever. That's not a gap in your team's execution. That's a structural problem with how GPU infrastructure gets connected. MaiaEdge fixes the structure.

Managed PBCs at your cluster edges, a cloud PCE across all your sites, and customers who get Layer 2 connectivity without ever touching a routing protocol. Your clusters stop being networking projects. They become product.